

VISHWANATH BIRADAR

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SKILLS

- **Embedded System**
- **Web Development**
- **C and C++ Programming**
- **Python Programming**
- **Basic VLSI**
- **Arduino and NodeMCU ESP8266 handheld experience**
- **Basic PIC Controller worked using MPLAB and Proteus (PIC16F877A)**

EDUCATION

- B.E. Electronics and Communication Engg. | BLDEA's V P Dr PG Halakatti College of Engineering & Technology, Vijayapura college CGPA: 7.65 | (2025)
- Diploma Electrical and Electronics Engg. | BLDE Association's Shree Sanganabasava Mahaswamiji Polytechnic, Vijayapura college CGPA: 7.8 | (2022)

ACADEMIC PROJECTS

- **Title: IOT-BASED COMMUNICATION SYSTEM FOR COLLEGE DEPARTMENT**
Role: Lead
Technology Stack: ESP8266, Python, MQTT
Project Link:
<https://github.com/vishwanathbiradar1/MQTT-Based-Communication-System.git>
Project description:
IoT-based communication system for the college department utilizing NodeMCU (ESP8266) and the MQTT protocol. The core innovation involved in-building the MQTT Broker and Wi-Fi Hotspot onto a single NodeMCU, which eliminated the need for external routers and solved critical IP address volatility challenges. This robust system, including a Smart Keypad Notification System that developed two-way ACK signal logic to ensure high data integrity, significantly improved communication efficiency and workflow, leading to high user satisfaction due to its reliability, real-time notifications, and cost-effective, scalable design.
- **Title: DETECTION AND CLASSIFICATION OF ACCOUSTIC ENVIRONMENT USING RANDOM FOREST AND NEURAL NETWORK**
Role: Lead
Technology Stack: Google Colab, Python, Laptop
Project Link:
<https://github.com/vishwanathbiradar1/DETECTION-AND-CLASSIFICATION-OF-ACCOUSTIC-ENVIRONMENT-USING-RANDOM-FOREST-AND-NEURAL-NETWORK.git>
Project description:
A real-time acoustic environment classification model was developed using Random Forest and Neural Network algorithms. MFCCs were utilized for feature extraction, and data was standardized with Standard Scaler before applying PCA for

dimensionality reduction. The models achieved 93% and 97% accuracy, respectively, and were evaluated using accuracy, confusion matrix, and precision, recall, and F1-score metrics. The deployed model is suitable for real-world applications such as hearing aids, smart homes, and automotive systems.

INTERNSHIP / TRAINING

- **EMBEDDED C PROGRAMMING | TRIOX Technology** (15 weeks, From OCT-2024 To MAR-2025)

I have completed internship on Embedded systems in the TRIOX Technology which is MSME (Micro, Small, and Medium Enterprises) registered company. I have learned Embedded C Programming, Proteus, MPLAB, Arduino IDE, etc. I also learned and had hands on experience on NodeMCU, Arduino, and making many projects and devices using the microcontrollers and electronics components and devices and sensors, etc and many more.

- **Full Stack Java with Web Development | The LearnHub** (From 25-10-2023 To 25-11-2023)

Excited to share my recent internship experience in Full Stack Java with Web Development! I dove into key technologies. I honed my skills building various projects, including clone websites, and gained hands-on experience in the full software development lifecycle. - Java, Spring Boot, REST APIs, MySQL, JavaScript, Bootstrap, HTML, CSS, Eclipse IDE, Visual Studio Code.

- **IoT and It's Application | Intra Institute Internship** (From 17-5-2023 To 3-6-2023)

Completed Intra Internship in IoT, gaining hands-on experience in designing and developing IoT-based systems and applications. BLDEA's V P Dr.P.G. Halakatti College of Engineering & Technology VIJAYAPURA. (Accredited by NBA & NAAC, Affiliated to VTU Belagavi).

- **Manufacturing of ACSR CONDUCTORS, DISTRIBUTIONS TRANSFORMERS & ALLIED MATERIALS | M/s. ANAND TRANSMISSION PRODUCTS PVT LTD** (From 02-05-2022 To 16-05-2022)

Gained hands-on experience in the manufacturing of ACSR conductors, distribution transformers, and allied materials at Anand Transmission Products Pvt. Ltd. Assisted in the production processes, quality control, and material handling within the manufacturing facility. Contributed to daily operations, adhering to safety protocols and industry standards. Collaborated with engineers and technicians to ensure efficient production and timely delivery of products.

ACHIEVEMENTS

- Awarded as "Best Project of the Year" during Open Day 2024-25 for presenting project work titled "Random Forest Based Detection and Classification of Acoustic Environment for Hearing Impaired Individuals".
- Participated in the IEEE INTERNATIONAL CONFERENCE ON INNOVATION & NOVELTY IN ENGINEERING AND TECHNOLOGY (INNOVA-2024) for Presenting a Poster titled "Detection and Classification of Acoustic Environment Using Random Forest and Neural Network".

HOBBY PROJECTS

- Traffic light with density-based control using Arduino uno and interface with sensor.
- Adaptive Home Lighting and Heating System.
- Multi-position contactless steering system with Battery boosting PV and TEG.

LANGUAGES

- English
- Hindi
- Kannada
- Marathi